

NIRF Analytics & Ranking Intelligence

Project Proposal and Progress Report

A data platform to understand the college's position in the NIRF Engineering ranking, the factors that drive it, and the likely outcome in 2026.

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Status: platform built and operational; one data enhancement in progress

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READING THIS DOCUMENT

Figures quoted for the college are **model estimates** unless stated otherwise, because NIRF publishes no score for institutions placed in rank bands. Every figure carries its source in grey.

Proposal

Why this work is needed, what it will deliver, and what it takes.

1 Executive summary

The National Institutional Ranking Framework (NIRF) is the Ministry of Education's annual ranking of Indian higher-education institutions. Saveetha Engineering College participates in the Engineering category every year. In 2024 and 2025 it was placed in the 201-300 band. For band placements NIRF publishes no score, so the college has no official view of how far it is from the next band, or which factors hold it back.

This proposal sets out a data platform that closes that gap. It collects every published NIRF result and the underlying data institutions submit, reconstructs the scoring method, estimates the college's own parameter scores, forecasts the 2026 outcome, and gives staff a web application to track current figures and test improvement scenarios. It is built entirely on free, open tools.

201-300 NIRF 2025 Engineering band (same in 2024)	≈ 40 Estimated score, 2026 filing	47 Forecast score at rank 100 in 2026	0 Recurring software cost; all free tiers
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Source: NIRF band pages (nirfindia.org); platform score model and 2026 forecast; resource plan in section 9.

2 Background: how NIRF ranks an institution

Each institution receives a score out of 100 built from five parameters. For Engineering the weights are:

Parameter	What it measures	Weight
TLR Teaching, Learning & Resources	Student strength, faculty-student ratio, faculty qualifications, spending per student	30%
RPC Research & Professional Practice	Publications, citations, patents, research funding and consultancy	30%
GO Graduation Outcomes	Placement and higher studies, graduation on time, median salary, PhDs awarded	20%
OI Outreach & Inclusivity	Students from other states, women, economically disadvantaged students, accessibility	10%
PR Perception	Survey of employers and academics	10%

Source: NIRF Engineering methodology 2025, nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf

Only the top 100 receive a numeric rank. Institutions below are grouped into bands (101-150, 151-200, 201-300) with no score. The formulas are published, but the normalisation step that converts raw figures into marks is not, so an institution cannot compute its own score from its own data.

3 Problem statement

- **No visibility of the college's own score.** NIRF has published no score for the college since 2019. Leadership sees only a band, not the distance to the next one.
- **No clear picture of what drives the result.** Without parameter scores it is not possible to tell whether teaching, research, outcomes or diversity is holding the college back.
- **The bar keeps rising.** The score at rank 100 rose from 41.93 (2023) to 43.95 (2024) to 45.55 (2025). Standing still means falling behind.
- **Research output is invisible in public data.** Publications and citations carry 75 of the 100 marks in the research parameter, yet NIRF's public data forms do not include them.
- **No internal tracking.** Current-year figures are compiled once a year for the submission, not monitored during the year when they can still be improved.

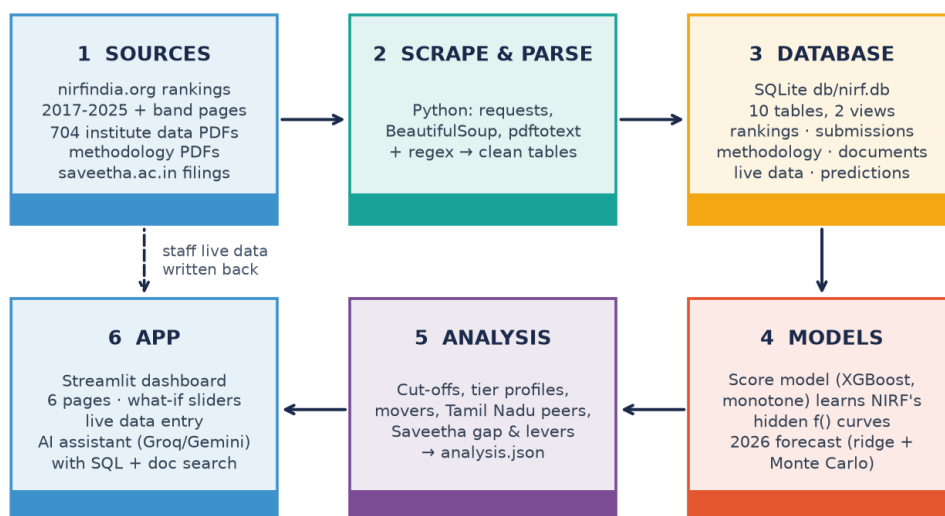
Source: NIRF Engineering ranking tables 2023-2025; NIRF methodology 2025.

4 Objectives

#	Objective	Measure of success
1	Build a complete, verifiable record of NIRF Engineering results	All years 2017-2025, checked against the live site
2	Reconstruct NIRF's scoring from public data	Published scores reproduced within a few points
3	Estimate the college's parameter scores	Estimate consistent with the band NIRF published
4	Identify the factors that separate the college from higher bands	Gap quantified per parameter
5	Forecast the 2026 outcome with honest uncertainty	Band probabilities, not a single guess
6	Give staff a tool to track figures and test scenarios	Web app usable without technical skills
7	Keep running costs at zero	Free and open-source components only

5 Proposed solution

An end-to-end analytics platform in six layers. Each layer is a script that can be re-run when NIRF publishes new results.



One command (`./pipeline.sh`) re-runs every step when NIRF 2026 is published

Layer	What it does
1 Data collection	Downloads NIRF ranking tables, band pages, applicant lists, 704 institution data forms, the methodology documents, and the college's own filings
2 Processing	Converts web pages and PDFs into clean, structured records (about 189 fields per institution)
3 Database	One file holding rankings, bands, submissions, methodology, documents, live staff data and forecasts
4 Models	Learns NIRF's hidden scoring step; estimates the college's scores; forecasts 2026
5 Analysis	Cut-offs, tier profiles, peer comparisons, the college's gap and improvement levers
6 Application	Six-page dashboard, what-if sliders, staff data entry, and a question-answering assistant

6 Methodology

- **Reconstructing the score.** For 700 institution-years both the submitted figures and the published parameter scores are known. A gradient-boosting model (XGBoost) learns the mapping for each parameter, with monotone constraints so that a better input can never lower a score.
- **Validation.** Each institution is held out and predicted from the others (grouped 5-fold cross-validation). The college's 2025 estimate is checked against the band NIRF actually published.
- **Forecasting.** The rank-100 cut-off is projected from its 2020-2025 trend. The college's 2026 score is simulated 20,000 times to reflect model error and the unknown perception score, giving band probabilities.
- **Scenario analysis.** Each improvement lever is applied to the college's filing one at a time and re-scored.
- **Data quality.** Every figure is traceable to a stored source file. Known site anomalies are corrected, and safety checks stop incomplete data from distorting results.

Source: `models/score_model.py`, `models/predict_2026.py` in the project repository.

7 Scope and deliverables

Deliverable	Description
Research database	NIRF results 2017-2025 and 704 parsed institution filings, in a single file
Scoring and forecast models	Parameter estimates for the college; 2026 band probabilities
Web application	Home, Top-100 Explorer, College Position, Gap & What-If, Prediction 2026, Live Data Entry, Ask the Data
Live data store	Hosted database so staff entries persist; local fallback when offline
Reports	This proposal and progress report; a staff analytics report; a handover document
Source code	Complete, version-controlled repository with a one-command refresh

Out of scope. Official NIRF scores for band placements (not published), the perception survey (not observable), and categories other than Engineering, which can be added later.

8 Implementation plan

Phase	Activity	Status
1	Study the NIRF methodology and weights	Complete
2	Collect rankings, bands, applicant lists and institution filings	Complete
3	Process and store the data; correct site anomalies	Complete
4	Build and validate the scoring model	Complete
5	Estimate the college's scores and quantify the gap	Complete
6	Forecast 2026 and model improvement scenarios	Complete
7	Build the web application and live data store	Complete
8	Reports, handover and source repository	Complete
9	Add peer publication data from an open research database	In progress
10	Collect the college's own publication and citation counts	Pending
11	Host the application online for staff	Pending
12	Extend to Innovation and SDG categories; yearly auto-refresh	Optional

9 Resource requirements

People

Role	Responsibility	Effort
Developer / analyst	Builds and maintains the platform, refreshes data yearly	Build complete; about 1 day a year to refresh
NIRF / IQAC coordinator	Enters current-year figures in the application	A few hours per term
Research office	Supplies publication and citation counts (Scopus / Web of Science)	Once a year

Technology and cost

Component	Used for	Cost
Python and open-source libraries	Data collection, processing, models, reports	Free
SQLite	Main database file	Free
Streamlit Community Cloud	Hosting the web application	Free
Supabase (free tier)	Storing staff-entered live data	Free
Groq and Google Gemini (free tiers)	Question-answering assistant	Free
OpenAlex	Open research-publication data for peer institutions	Free (daily request limit)
GitHub	Source code and version history	Free
Hardware	Any standard laptop	Existing

Total recurring cost: **nil**. No card or paid plan is needed for any component.

Risk	Impact	Mitigation
Estimates mistaken for official NIRF scores	Misleading decisions	Every figure labelled as an estimate; uncertainty shown as ranges
NIRF changes its methodology	Forecast becomes outdated	Methodology re-read each year; one command refreshes everything
Research output not visible in public data	Research estimate least certain	College enters its own counts; safety check blocks unreliable peer data
Open research database covers smaller colleges poorly	Biased research model	Coverage checked by rank band before the data is used
Free-tier limits (request quotas, database pausing)	Temporary interruption	Local fallback storage; resumable data collection
Credentials exposed	Security	Keys kept in a private settings file excluded from the repository

- All reference data is public: NIRF results, published institution filings and the college's own public filings.
- Staff entries are stored with the name of the person and the date, so every change is traceable.
- No personal student data is collected. Faculty names appear only where the college itself published them.
- API keys and the database password are held in a private file that is never committed or shared.

PART B

Progress report

What has been built, what the data shows, and what remains.

12 Work completed

Area	Result
Rankings collected	4,001 scored entries across Engineering, Overall, College and University, 2017-2025
Band placements	3,902 entries for 101-150, 151-200 and 201-300
Applicant records	47,173 entries
Institution filings parsed	704 PDFs, about 189 fields each
Documents indexed for the assistant	713
Scoring model accuracy	Published totals reproduced within 2.05 points on average
Forecast model check	2025 predicted from 2024 with average error 1.36 points
Web application	Seven pages, tested; live data saved to a hosted database
Data corrections	2016 duplicate pages removed; mislabelled band pages fixed; a missing 2021 placement recovered

Source: Project database db/nirf.db; model_report.json; prediction_2026.json.

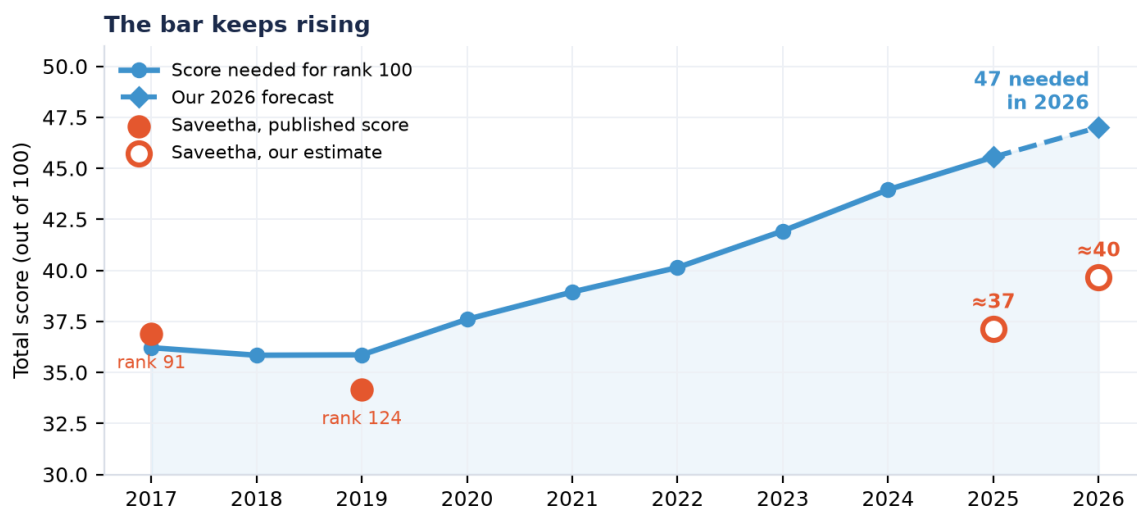
13 Findings

13.1 The college's NIRF record

Year	Result	Score
2017	Rank 91	36.88
2018	Band 101-150	—
2019	Rank 124	34.14
2020	Band 201-250	—
2021	Band 201-250	—
2022	Applied; not placed in a published list	—
2023	Applied; not placed in a published list	—
2024	Band 201-300	—
2025	Band 201-300	—

Source: NIRF Engineering ranking and band pages, verified against nirfindia.org. A dash means no score was published.

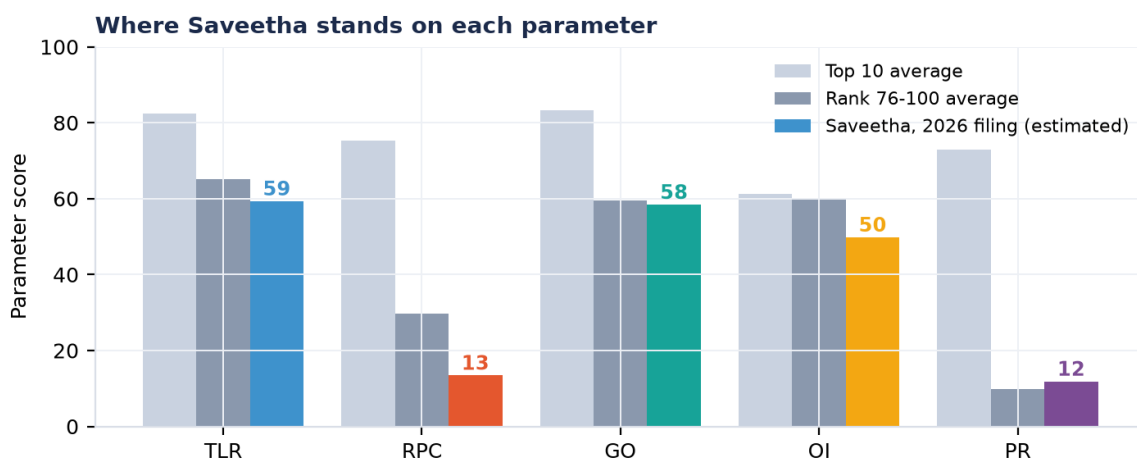
13.2 The rising bar



Source: NIRF ranking tables (cut-off); score model (college estimates, open circles).

The score at rank 100 has risen about 1.6 points a year since 2020. The college's filings are estimated at 37.1 (2025) and 39.7 (2026): a real improvement, but the bar is moving almost as fast.

13.3 Where the gap is



Source: Score model estimates from the 2026 filing; averages of published 2025 scores.

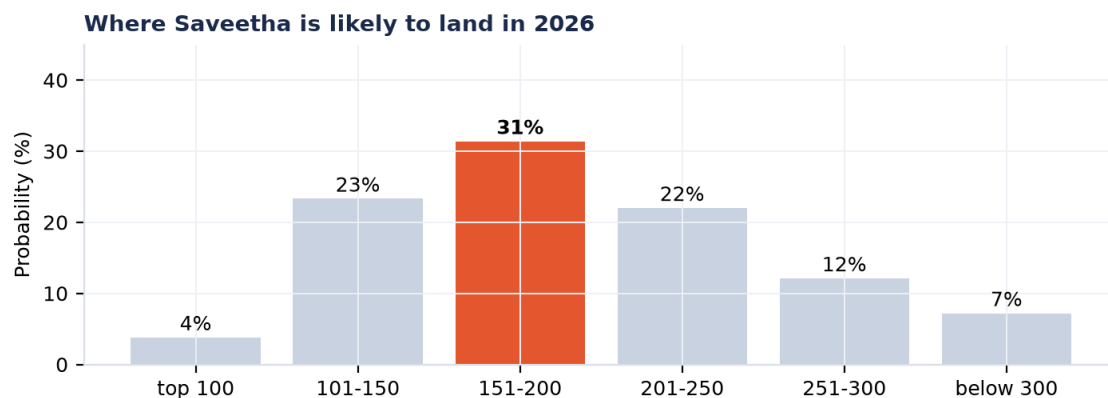
Parameter	Estimated points lost versus rank 90-100
RPC Research	+4.11
TLR Teaching & resources	+1.56
OI Outreach	+0.78
GO Graduation outcomes	+0.46
PR Perception	-0.11

Source: analysis.json (computed from the database).

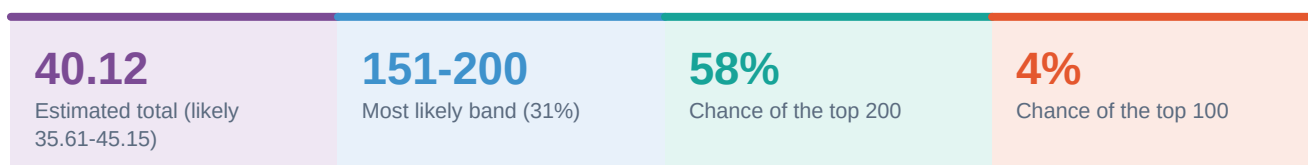
KEY FINDING

Research accounts for about 4.1 of the weighted points separating the college from the rank 90-100 group. Placements, graduation on time and women's participation are already at or above that group.

13.4 2026 outlook

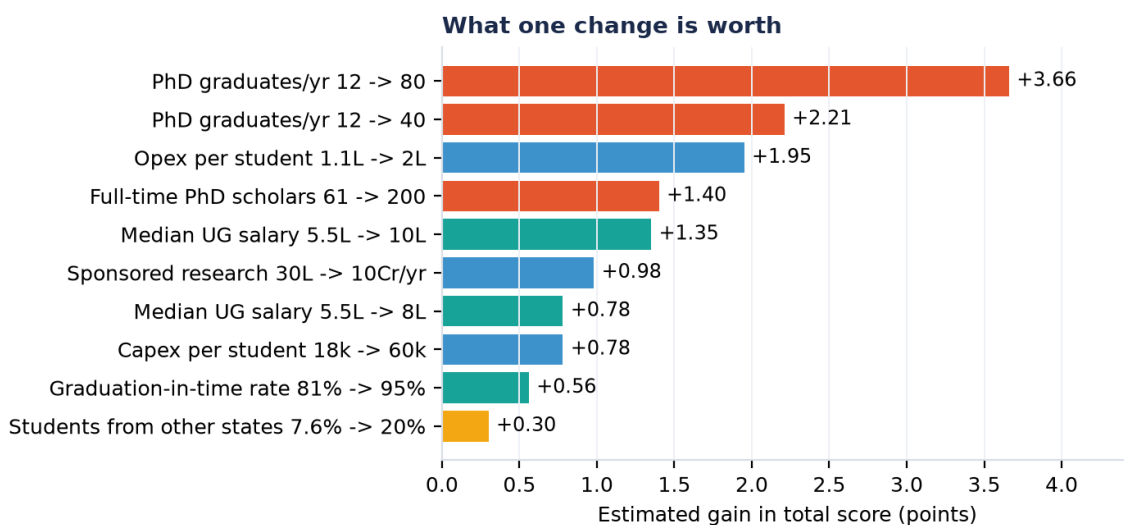


Source: prediction_2026.json: 20,000 simulations of the 2026 filing.



The adjacent bands are close (101-150 at 23%, 201-250 at 22%), so the result is best read as "the 100s to low 200s". The range is wide because perception and publication counts cannot be observed.

14 Recommendations



Source: prediction_2026.json: each change applied alone to the 2026 filing and re-scored.

- **Expand the PhD pipeline.** PhDs awarded is the largest single lever. Enrolment decisions now determine awards three to four years later.
- **Grow publications and citations.** They are 75 of the 100 research marks. Track Scopus output per department and avoid retractions, which NIRF now penalises.
- **Raise spending per student** on laboratories, equipment and e-resources.
- **Lift the median graduate salary**, which depends on the middle of the batch rather than top offers.
- **Increase sponsored research and admissions from other states.**
- **Maintain current strengths** in placement, graduation on time and inclusion.

15 Pending items and next steps

Item	Status	Detail	Owner
Peer publication data	In progress	144 of 277 institutions matched so far. Smaller colleges are poorly covered, so the data stays switched off until coverage is representative.	Developer
College publication counts	Pending	Papers and citations for the last three years. The college is not listed in the open database, so this must come from the research office.	Research office
Online hosting	Pending	Publish on Streamlit Community Cloud so staff can use it from any device.	Developer
Staff trial	Pending	Coordinator enters current figures and reviews each page.	NIRF coordinator
Further categories and auto-refresh	Optional	Innovation and SDG categories; refresh when NIRF 2026 is published.	Developer

16 Limitations

- The college's parameter scores are estimates; NIRF has published none since 2019.
- Research carries most of the uncertainty. Model accuracy for research is R^2 0.78, against 0.82 for graduation outcomes.
- Perception is a survey and is treated as an assumption, not an estimate.
- The forecast assumes NIRF keeps its 2025 methodology.

Parameter	Institutions used	Variation explained (R^2)	Typical error
TLR	700	0.73	4.3 points
RPC	700	0.78	7.5 points
GO	700	0.82	3.7 points
OI	700	0.74	3.0 points

Source: model_report.json; grouped 5-fold cross-validation.

PART C

Resources and references

Where every figure comes from, and the tools used.

17 Data sources

Source	Content	Address
NIRF ranking tables	Scores and ranks, 2017-2025	<a href="http://nirfindia.org/Rankings/<year>/EngineeringRanking.html">nirfindia.org/Rankings/<year>/EngineeringRanking.html
NIRF band pages	Band placements without scores	<a href="http://nirfindia.org/Rankings/<year>/EngineeringRanking150.html">nirfindia.org/Rankings/<year>/EngineeringRanking150.html (...200, ...300)
NIRF applicant lists	All participating institutions	<a href="http://nirfindia.org/Rankings/<year>/EngineeringRankingALL.html">nirfindia.org/Rankings/<year>/EngineeringRankingALL.html
Institution data forms	Submitted figures of ranked institutions	<a href="http://nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<id>.pdf">nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<id>.pdf
NIRF methodology	Parameters, marks, weights, formulas	nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf
College filings	NIRF 2025 and 2026 submissions	saveetha.ac.in (NIRF documents)
OpenAlex	Publication and citation counts of peer institutions	openalex.org

Copies of every downloaded page and document are stored unchanged in the repository under `data/raw`, so any figure can be checked.

18 Tools and platform

Area	Tools
Programming	Python 3.13
Collection and parsing	requests, BeautifulSoup, poppler pdftotext
Data and storage	pandas, SQLite, PostgreSQL (Supabase)
Modelling	XGBoost, scikit-learn
Application and charts	Streamlit, Plotly, Matplotlib
Reports	ReportLab
Assistant	Groq and Google Gemini language models; BM25 document search
Version control	Git and GitHub

19 Project links

Item	Location
Source code	https://github.com/rnandakishore2408/nirf_analytics
Application (local)	http://localhost:8501 after running <code>./run.sh</code>
Application (online)	To be published on Streamlit Community Cloud
Handover document	<code>docs/HANDOVER.md</code> in the repository
Staff analytics report	<code>docs/NIRF_Analytics_Report.pdf</code> in the repository
Refresh all data and models	<code>./pipeline.sh</code>

Term	Meaning
NIRF	National Institutional Ranking Framework, Ministry of Education
Band	A range such as 201-300 used instead of a rank below 100; no score is published
Cut-off	The lowest score inside the top 100 in a given year
Filing	The data form an institution submits to NIRF, published as a PDF
Estimate	A score computed by the platform's model, not by NIRF
R ²	Share of variation a model explains; 1.0 is perfect
Cross-validation	Testing a model on institutions it was not trained on
Monte Carlo simulation	Repeating a calculation many times with random variation to obtain a range
OpenAlex	A free, open index of research publications and institutions